JESSE MUUYA NDAGU

Full-Stack Developer · Cloud & Application Security

Nairobi, Kenya | +254 746 272 251 | jessemuuya555@gmail.com
linkedin.com/in/jesse-muuya-5285842b6 | cal-me-j.github.io | github.com/cal-me-j

SUMMARY

Full-stack developer (React, TypeScript, NestJS, PostgreSQL) building and operating multi-tenant production systems with M-Pesa, Paystack and Flutterwave payments. I deploy on AWS EC2 with Docker, Nginx, Redis and CI/CD, and build security in: tenant isolation, database-enforced access control, signed webhooks and rate limiting. Growing toward cloud and AI application security.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, PHP

Frontend: React, Next.js, Vite, Progressive Web Apps, Tailwind CSS

Backend: NestJS, Fastify, Node.js, Express, Prisma, REST APIs, Supabase

Data: PostgreSQL (Neon, Supabase), MongoDB, Redis

Cloud & DevOps: AWS EC2, AWS IVS, Docker, Nginx, GitHub Actions, GHCR, Cloudflare (DNS, R2, Workers), Let's Encrypt, Vercel, Render

Security: Postgres RLS, RBAC, tenant isolation, HMAC-signed tokens, webhook signature verification, rate limiting, Helmet

Integrations: M-Pesa Daraja, Paystack, Flutterwave, YouTube API, Resend, SendGrid

PROJECTS

ObinnaTV Studios CMS Platform

Sole developer

React / Next.js, NestJS (Fastify), Prisma, PostgreSQL (Neon), AWS IVS

- Multi-tenant content platform: site builder, YouTube-synced video and blogs, free and paid live streams, promo codes, and ticketing with QR gate check-in.
- Deployed on AWS EC2: three load-balanced API containers behind Nginx, images built by GitHub Actions on ARM runners and shipped via GHCR, Let's Encrypt TLS. Migrated web and admin from Vercel with a planned DNS cutover and rollback.
- Moved media from Vercel Blob to Cloudflare R2 with tenant-scoped upload keys and an automated orphan-cleanup job.
- Security: per-route rate limiting on shared Redis counters across replicas, Helmet, payload limits, signature-verified Paystack and Flutterwave webhooks, tenant scoping at the Prisma layer, and HMAC-signed single-use QR tickets with atomic check-in.

Baobab – Multi-Tenant POS / ERP / CRM

Sole developer

NestJS, Prisma, PostgreSQL, React

- Modular monolith with hard module boundaries covering POS, inventory, purchasing, invoicing, CRM and HR, built on an immutable double-entry ledger.
- Tenant-scoped identity with role-based permissions; M-Pesa payment flows; eTIMS e-invoicing integration built and awaiting KRA device registration.

Road Works ERP

Sole developer

React / Vite, Supabase (PostgreSQL), PWA, Cloudflare Workers

- Extended a client prototype into a working system. Replaced the prototype's backend with Supabase Auth and Postgres, and migrated the data layer.
- Customisable roles and permissions enforced in the database through Row Level Security and SECURITY DEFINER functions, with tenant-scoped policies and approval rights held as a separate permission.
- Append-only fuel logs with correction records, and fuel-cycle tracking that flags inconsistent consumption.

African Payment Gateway Engine

Open product

TypeScript, Node.js, Prisma, Zod

- Framework-agnostic engine for M-Pesa STK Push, Paystack and Flutterwave: signed webhooks, runtime payload validation, phone normalisation and cron-driven reconciliation of stuck transactions.

Other projects

- Methodist Church Membership System (React, Node.js/Express, MongoDB Atlas): registration, attendance, records and reporting.
- Political Intelligence System and Projects Tracker (React/Vite, Supabase, PWA); Ticketing App (Next.js, M-Pesa Daraja).

EXPERIENCE

Software Developer *Feb 2026 – Present*
Starling Technologies Limited, Nairobi

- Full-stack development of multi-tenant web platforms; see Projects.

Freelance Full-Stack Developer *Oct 2025 – Feb 2026*
AttachMe (attachme.co.ke)

- Built and deployed the internship-placement platform end to end: React front end, Node.js back end with REST APIs, MongoDB, hosting and deployment.
- Applied modern UI practices with Expo and Gluestack UI.

IT Technician & Developer *May 2025 – Aug 2025*
Robisearch Africa Limited

- Installed and configured POS, inventory management, biometric attendance and CCTV systems; configured networking for printers and workstations.
- Diagnosed hardware and software faults, migrated client data, trained users, and assisted with company websites.

CERTIFICATIONS

IBM Cybersecurity Fundamentals (IBM SkillsBuild, Mar 2026) | IBM Cloud Computing Fundamentals (IBM SkillsBuild, Mar 2026)

Building LLM-based AI Applications (ATF AI School, Africa Technology Foundation, Oct 2026)

EDUCATION

BSc Computer Science *2019 – 2025*
Presbyterian University of East Africa, Thogoto – Second Class Honours (Lower Division)